

Classroom



In this section of *Resonance*, we invite readers to pose questions likely to be raised in a classroom situation. We may suggest strategies for dealing with them, or invite responses, or both. “Classroom” is equally a forum for raising broader issues and sharing personal experiences and viewpoints on matters related to teaching and learning science.

Rational Points on Circles*

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A primitive Pythagorean triple is a triple of positive integers (a, b, c) such that they are pairwise coprime, and $a^2 + b^2 = c^2$. Geometrically, it gives a rational point $(\frac{a}{c}, \frac{b}{c})$ on the unit circle $x^2 + y^2 = 1$. In other words, $1 = x^2 + y^2$ has rational solutions (in fact, these are infinitely many and are characterized easily). There are other positive integers n for which $n = x^2 + y^2$ has rational solutions. For instance,

$$\left(\frac{1}{5}\right)^2 + \left(\frac{7}{5}\right)^2 = 2,$$

$$\left(\frac{1}{5}\right)^2 + \left(\frac{18}{5}\right)^2 = 13,$$

etc. Evidently, if $n = x^2 + y^2$ has a solution in rational numbers x, y , then $nt^2 = X^2 + Y^2$ also has rational solutions for any rational number t . In this note, I determine all rational numbers α for which $\alpha = x^2 + y^2$ has rational solutions. In other words, I find all rational $\alpha > 0$ for which the circle of radius α centered at 0 has rational points. The problem can also be viewed as a question on “norms” as I will explain in this article.

Keywords

Sum of two squares, Pythagorean triple, norm maps, rational points on circle, Hilbert’s theorem, Gauss circle problem.

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1. Rational Points of $x^2 + y^2 = t$ for $t \in \mathbb{Q}$

The result I prove is as follows:

Proposition. *A positive rational number t is a sum of squares of two rational numbers if, and only if, $t = nu^2$, $u \in \mathbb{Q}$, where n is a square-free natural number that is not divisible by any prime of the form $4k + 3$.*

Proof. Suppose a rational number $t = \frac{r}{s}$, in its lowest terms, is expressible as $\frac{r}{s} = x^2 + y^2$, with $x, y \in \mathbb{Q}$. There exists a non-square positive integer n such that rs can be written as $rs = v^2n$ for some natural number v . Then, since

$$n = \frac{rs}{v^2} = \left(\frac{x}{v}\right)^2 + \left(\frac{y}{v}\right)^2,$$

I deduce that

$$t = \frac{rs}{s^2} = \left(\frac{v}{s}\right)^2 n,$$

with n a square-free positive integer that is a sum of two squares of rational numbers.

Conversely, if n is a square-free positive integer that is a sum of two squares of rational numbers, evidently nu^2 is also similarly expressible, where u is any rational number. Now, suppose n is a square-free positive integer such that

$$n = \left(\frac{a}{b}\right)^2 + \left(\frac{c}{d}\right)^2,$$

where $\gcd(a, b) = 1 = \gcd(c, d)$. Then, $a^2d^2 + b^2c^2 = nb^2d^2$. Thus, $b^2|d^2$ as it divides the second and third terms above, and $\gcd(b^2, a^2) = 1$. Similarly, $d^2|b^2$ as $\gcd(d^2, c^2) = 1$. So $d^2 = b^2$, which gives

$$n = \left(\frac{a}{b}\right)^2 + \left(\frac{c}{d}\right)^2,$$

that is,

$$nb^2 = a^2 + c^2.$$

Here, $\gcd(a, b) = 1 = \gcd(c, b)$ and n is square-free. It is well known (but I recall the argument briefly for the sake of completion) that this happens if, and only if, no prime of the form $4k + 3$



divides n . Indeed, if a prime p of the form $4k + 3$ divides n , then p does not divide either of a and c ; otherwise, p would divide both a and c , and then p^2 would divide the RHS, and hence p would divide b^2 , a contradiction to the fact that b is coprime to a, c . Thus, a^2 is congruent to $-c^2$ modulo p , which would imply that -1 is a square modulo p , an impossibility. Therefore, no prime of the form $4k + 3$ divides n . Conversely, it is a well-known fact due to Fermat that primes of the form $4k + 1$ are sums of two squares of positive integers. Of course $2 = 1^2 + 1^2$ as well. Finally, if $k = p^2 + q^2, \ell = r^2 + s^2$ in integers, then

$$k\ell = (pr + qs)^2 + (ps - qr)^2.$$

Therefore, I have proved the proposition.

2. Rephrasing in Terms of Norm Maps

Here, for a square-free integer d – positive or negative, I look at quadratic fields $\mathbb{Q}[\sqrt{d}] = \{x + y\sqrt{d} | x, y \in \mathbb{Q}\}$. The norm map N of that field is defined as $N : \mathbb{Q}[\sqrt{d}] \rightarrow \mathbb{Q}$

$$N(x + y\sqrt{d}) = x^2 - dy^2.$$

It has the property that it respects multiplication on both sides. For example, consider $d = -1$ in which case the norm map is

$$N(x + iy) = x^2 + y^2.$$

Thus, note that a positive rational number t is a sum of two rational squares if, and only if, it is a norm of an element from the field $\mathbb{Q}[\sqrt{d}]$. This point of view is useful; for instance, I may determine all Pythagorean triples as well as more general integer triples that are sides of certain triangles, as follows.

The so-called Theorem 90 of Hilbert implies that any element $x + y\sqrt{d}$ of norm equal to 1 is expressible as

$$x + y\sqrt{d} = \frac{u + v\sqrt{d}}{v - \sqrt{d}v},$$



for some $u, v \in \mathbb{Z}$ (the theorem implies this assertion with u, v rational but one can cancel the common denominator). If I apply Hilbert's Theorem 90 to $K = \mathbb{Q}[\sqrt{d}]$, I get that for any element $(x + y\sqrt{d}) \in \mathbb{Q}[\sqrt{d}]$ (i.e., $x, y \in \mathbb{Q}$) with norm $(x^2 - dy^2) = 1$, $\exists u, v \in \mathbb{Q} \ni (x + y\sqrt{d}) = \frac{(u+v\sqrt{d})}{(u-v\sqrt{d})}$, as, $\frac{(u+v\sqrt{d})}{(u-v\sqrt{d})} = \frac{(u^2+v^2+2uv\sqrt{d})}{(u^2-dv^2)} = \frac{(u^2+v^2)}{(u^2-dv^2)} + \frac{2uv\sqrt{d}}{(u^2-dv^2)}$ (i.e., $x = \frac{(u^2+v^2)}{(u^2-dv^2)}, y = \frac{2uv}{(u^2-dv^2)}$).

Now applying Hilbert's Theorem 90 to $\mathbb{Q}[i]$ I get for any element, $(x + iy) \in \mathbb{Q}[i]$, with norm $(x^2 + y^2) = 1$, $\exists u, v \in \mathbb{Z} \ni (x + iy) = \frac{(u+iv)}{(u-iv)} = \frac{(u^2-v^2)}{(u^2+v^2)} + \frac{2uvi}{(u^2+v^2)}$. So, here I get Pythagorean triplets of integers a, b, c , where $a = (u^2 - v^2), b = 2uv, c = (u^2 + v^2)$.

The same argument can be applied to other triangles where an angle has rational cosine and sine – see [1] mentioned at the end. For instance, I recall the following parametrization from that reference.

Take an integer-sided $\triangle ABC$ with $\cos \angle C = \frac{p}{q}$, sides $a, b, c \in \mathbb{N}$ opposite to $\angle A, \angle B, \angle C$, respectively. Take $x = \frac{a}{c}, y = \frac{b}{c} \in \mathbb{Q}$.

Now, $(a^2 + b^2 - c^2) = 2ab \cos \angle C \Rightarrow x^2 + y^2 - 2xy\frac{p}{q} = 1$. By Hilbert's Theorem 90,

$$\begin{aligned} \exists u, v \in \mathbb{Q} \ni \left(x - y \frac{(p - \sqrt{p^2 - q^2})}{q} \right) &= \frac{u + v \sqrt{p^2 - q^2}}{u - v \sqrt{p^2 - q^2}} \\ \Rightarrow \frac{a}{c} - \frac{(p - \sqrt{p^2 - q^2})}{q} \frac{b}{c} &= \frac{u^2 + v^2 (p^2 - q^2) + 2uv \sqrt{p^2 - q^2}}{u^2 - v^2 (p^2 - q^2)} \\ \Rightarrow \frac{(aq - bp)}{qc} &= \frac{u^2 + v^2 (p^2 - q^2)}{u^2 - v^2 (p^2 - q^2)}, \frac{b}{qc} = \frac{2uv}{u^2 - v^2 (p^2 - q^2)} \\ \Rightarrow \frac{b}{c} &= \frac{2uvq}{u^2 - v^2 (p^2 - q^2)}, \frac{a}{c} = \frac{u^2 + v^2 (p^2 - q^2) + 2uvp}{u^2 - v^2 (p^2 - q^2)}. \end{aligned}$$

So, $(a : b : c) = (u^2 + v^2(p^2 - q^2) + 2uvp : 2uvq : u^2 - v^2(p^2 - q^2))$ gives a parametrization of sides of the integer-sided triangle.

I end this section with a subtle question involving norms. Note that -1 is the norm of $\left(\frac{3}{5} + \frac{\sqrt{34}}{5}\right) \in \mathbb{Q}[\sqrt{34}]$.



But there is no “integral” element $a + b\sqrt{34}$ (i.e., with $a, b \in \mathbb{Z}$) such that $-1 = a^2 - 34b^2$. It is in general a subtle question as to when -1 is a unit of an “integral” element from a quadratic field – see [2].

3. Schinzel and Gauss on Circles

I end with a result that is not directly related to the problem I discussed in this note, but it is an interesting result on lattice points on circles. This was observed by A Schinzel:

For any positive integer n , there exists a circle C whose circumference contains exactly n lattice points. This is defined as:

$$C = \begin{cases} \left(x - \frac{1}{2}\right)^2 + y^2 = \frac{5^{\frac{(n-2)}{2}}}{4} & n = 2k, \\ \left(x - \frac{1}{3}\right)^2 + y^2 = \frac{5^{(n-1)}}{9} & n = 2k + 1 \end{cases} \quad \exists k \in \mathbb{N}.$$

In this regard, it is relevant to mention the famous Gauss’s circle problem. Gauss showed that the number $N(r)$ of lattice points inside a circle of radius r with origin as center satisfies the property $N(r) = \pi r^2 + e(r)$, where $e(r) \leq 2\sqrt{2}\pi r$.

The motivation for expecting this is that the number of lattice points inside the circle is roughly the area inside a circle because each unit square contains one lattice point. Now, the question is, what the best power $\theta < 1$ is, for which $\exists$ a constant ‘ C ’ such that $N(r) \leq Cr^\theta$. This is an active area of research even now, and the best-known bound is approximately 0.63. It is conjectured that the best bound one can expect is any constant $> \frac{1}{2}$. This is one of the old unsolved problems in number theory.

Suggested Reading

- [1] B Sury, 90 provides the right angle of approach, *Parabola*, Vol.59, No.2, pp.2–4, 2023.
- [2] Fouvery and Kluners, ‘On the negative Pell equation, *Anna. Math.*, Vol.172, No.3, pp.2035–2036, 2010.

